

Algorithms Problem Solving

Level 2

Digit Frequency

The background of the slide is a photograph of a modern building's interior. It features a large, curved glass dome ceiling that allows natural light to filter through. In the foreground, the silhouettes of several people are visible, some standing and others sitting, suggesting a collaborative workspace or a public area. The overall color palette is muted, with greys and blues, which makes the white and yellow text stand out.

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Definition

- ◆ Digit frequency means counting how many times each digit from `0` to `9` appears in a number.
- ◆ This program reads one number and prints the frequency of every digit that appears at least once.

Explanation

- ◆ ``ReadNumber()`` reads a non-negative number from the user.
- ◆ ``CountFrequency()`` counts how many times a specific digit appears in the given number.
- ◆ ``N % 10`` extracts the last digit.
- ◆ ``N = N / 10`` removes the last digit.
- ◆ If the extracted digit equals the target digit, ``freq`` is increased.
- ◆ ``PrintResult()`` uses a ``for`` loop from ``0`` to ``9``.
- ◆ For every digit, it calls ``CountFrequency()`` to find how many times that digit appears.
- ◆ ``if (freq > 0)`` makes sure that only digits that actually appear in the number are printed.

Steps in main()

1. Read the main number.
2. Start a loop from digit `0` to digit `9`.
3. Count the frequency of the current digit.
4. Extract each digit from the number using `% 10`.
5. Remove each digit using `/ 10`.
6. Increase the frequency when a match is found.
7. Print the digit if its frequency is greater than `0`.
8. Continue until all digits from `0` to `9` have been checked.

💡 Example

◆ Input:

`122342`

◆ The program checks each digit:

`0 → 0 times`

`1 → 1 time`

`2 → 3 times`

`3 → 1 time`

`4 → 1 time`

`5 → 0 times`

`6 → 0 times`

`7 → 0 times`

`8 → 0 times`

`9 → 0 times`

➡ Result

`The digit 1 frequency is 1 Time(s).`

`The digit 2 frequency is 3 Time(s).`

`The digit 3 frequency is 1 Time(s).`

`The digit 4 frequency is 1 Time(s).`

Note

- ◆ The `for` loop checks all possible digits from `0` to `9`.
- ◆ `CountFrequency()` is reused for every digit instead of writing separate code for each digit.
- ◆ `if (freq > 0)` prevents digits with zero occurrences from being printed.

⊘ Warning

- ◆ `CountFrequency()` receives `Digit` as a `float`, but a digit should normally be represented by `int`.
- ◆ The program does not count a leading zero because leading zeros are not stored as part of an integer. For example, `00123` is read as `123`.



Comparison

▨ `#8\_Digit\_Frequency`

- ✓ Searches for one specific digit.
- ✓ Example: `122342`, digit `2` → `3 times`.

▤ `#9\_Digit\_Frequency`

- ✓ Checks all digits from `0` to `9`.
- ✓ Example: `122342` → finds frequencies for `1`, `2`, `3`, and `4`.

✦ Summary

- ✓ Check every digit from `0` to `9`.
- ✓ Use `CountFrequency()` to count occurrences.
- ✓ `% 10` extracts the last digit.
- ✓ `/ 10` removes the last digit.
- ✓ Print only digits whose frequency is greater than `0`.
- ✓ `122342` contains `1` once, `2` three times, `3` once, and `4` once.

Thank you 😊